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V810i S2EX Confirmation, Diagnostics and Adjustments (CD&A)





Automated Board Inspection

Confirmation, Diagnostics & Adjustments

Proper and periodic confirmation and adjustment of the V810 as described below allows a user to achieve the desired measurement repeat-ability. Therefore it is unnecessary at this time for the V810 to be calibrated using traceable instrumentation or any other physical standard to achieve its desired results. Due to this reason, V810 Advanced X-Ray Inspection System does not require any periodic calibration as in measurement equipment. This reduces the downtime for calibration which could take days.

An automatic periodic checking will be carried out at an interval time to make sure the system condition is within intended specification. And if found not within intended specification, the software will adjust its calculation to make sure the inspection performance is consistent with previous inspections. Until a point that calculation adjustment is impossible, the machine will prompt message stating the machine needs to be serviced.

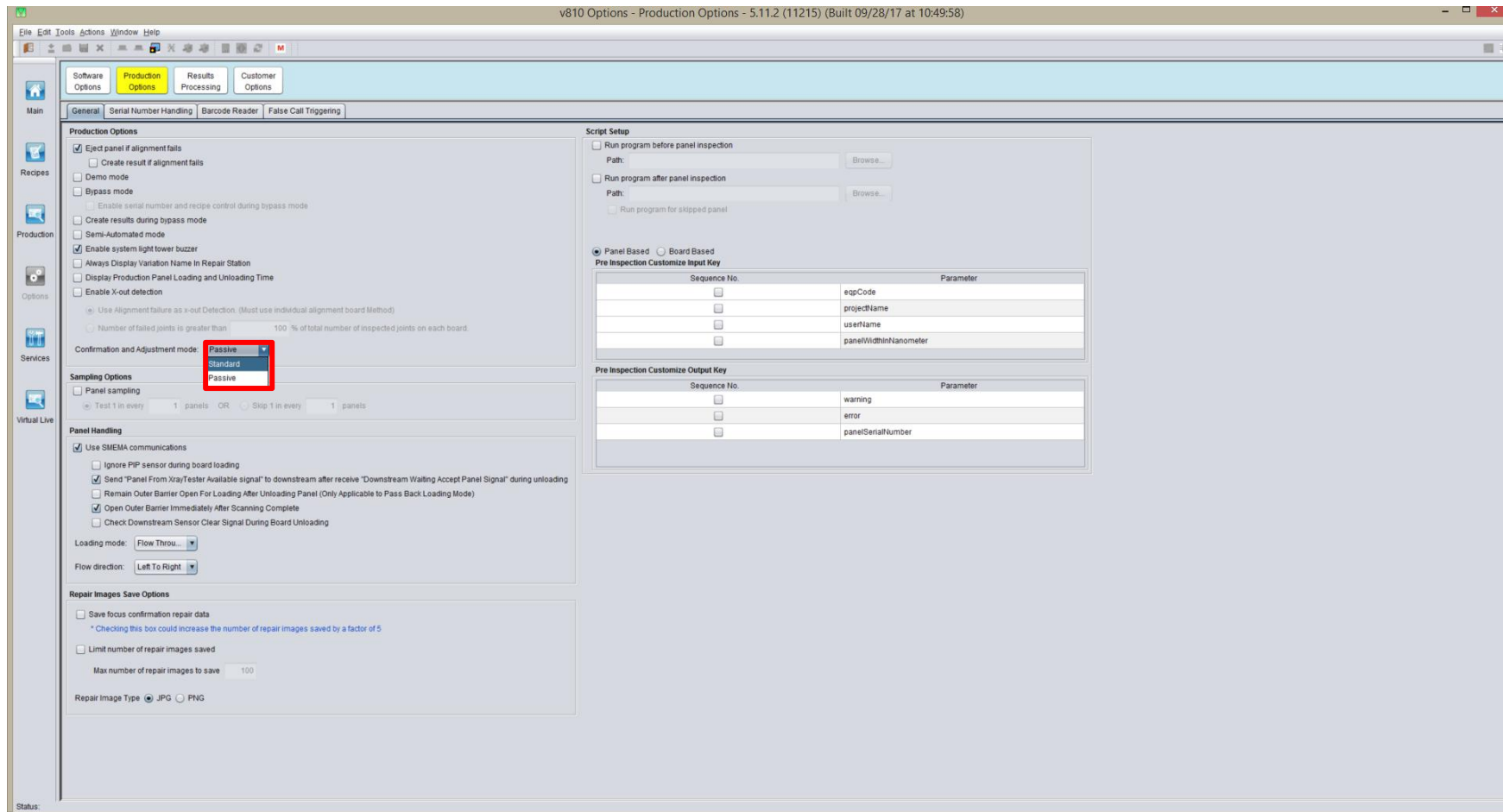
Confirmation, Diagnostics and Adjustment

Vitrox V810 confirmation and adjustments are used to compensate for minor changes in the system hardware that occur during normal use of the system during its lifetime, and needed to maintain optimal system performance over time. Variation in environmental conditions may also indicate that these procedures should be completed.

Confirmation, diagnostic and adjustments must be in the following sequence:

- System Communication Network Confirmation
- Area Mode Image Confirmation
- Camera Grayscale Adjustment
- Camera Adjustment Confirmation
- Camera Dark Confirmation
- Camera Light Confirmation
- X-ray Spot Adjustment
- Stage Hysteresis Adjustment
- System Offsets Adjustment
- Magnification Adjustment
- Base Image Quality Confirmation
- System Image Quality Confirmation
- X-ray Source Confirmation

V810 software will automatically execute each of the tasks above at their respective timeout values . The timeout values are based on the rate of changes that may cause the machine out of specification over time. Details on the operation of all of the confirmation and adjustment functions can be found in one of the ViTrox Service Guide named “Confirmation, Diagnostics & Adjustment (CD&A)”.



- Standard mode, system will run CD&A periodic timeout task for both magnification (Low and high) when needed
- Passive mode, system will run CD&A periodic timeout task accordingly to selected recipes setting (low, high or both)

v810 Service - Confirmation, Diagnostics and Adjustments - 5.11.2 (11215) (Built 09/28/17 at 10:49:58)

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Test List:

- Service Folder
 - Scheduled Tasks
 - System Communication Network Confirmation
 - Area Mode Image Confirmation
 - Camera Grayscale Adjustment
 - Camera Adjustment Confirmation
 - Camera Dark Confirmation
 - Camera Light Confirmation
 - X-ray Spot Adjustment
 - Stage Hysteresis Adjustment
 - System Offsets Adjustment
 - Magnification Adjustment
 - Base Image Quality Confirmation
 - System Image Quality Confirmation
 - High Mag Camera Grayscale Adjustment
 - High Mag Camera Adjustment Confirmation
 - High Mag Camera Dark Confirmation
 - High Mag Camera Light Confirmation
 - High Mag X-ray Spot Adjustment
 - High Mag Stage Hysteresis Adjustment
 - High Mag System Offsets Adjustment
 - High Magnification Adjustment
 - High Mag Base Image Quality Confirmation
 - High Mag System Image Quality Confirmation
 - X-ray Source Confirmation
 - Manual Tasks
 - Manual Confirmations
 - X-ray Z Axis Motor Motion Confirmation
 - Panel Handling Confirmation
 - Motion Repeatability Confirmation
 - Reverse X-ray Spot Image Confirmation
 - Confirm Camera Calibration Images
 - High Mag Confirm Camera Calibration Images
 - Manual Optical Adjustments
 - Optical System Fiducial Adjustment
 - Optical Adjustment
 - Optical System Offset Adjustment
 - Manual Optical Confirmations
 - Front Optical Measurement Repeatability Confirmation
 - Rear Optical Measurement Repeatability Confirmation

Run while system start-up, time-out or manual trigger

Motion repeat-ability task, manually trigger. # Panel Handling confirmation will need manual load panel into system before run

PSP module, hardware adjustment and repeat-ability confirmation

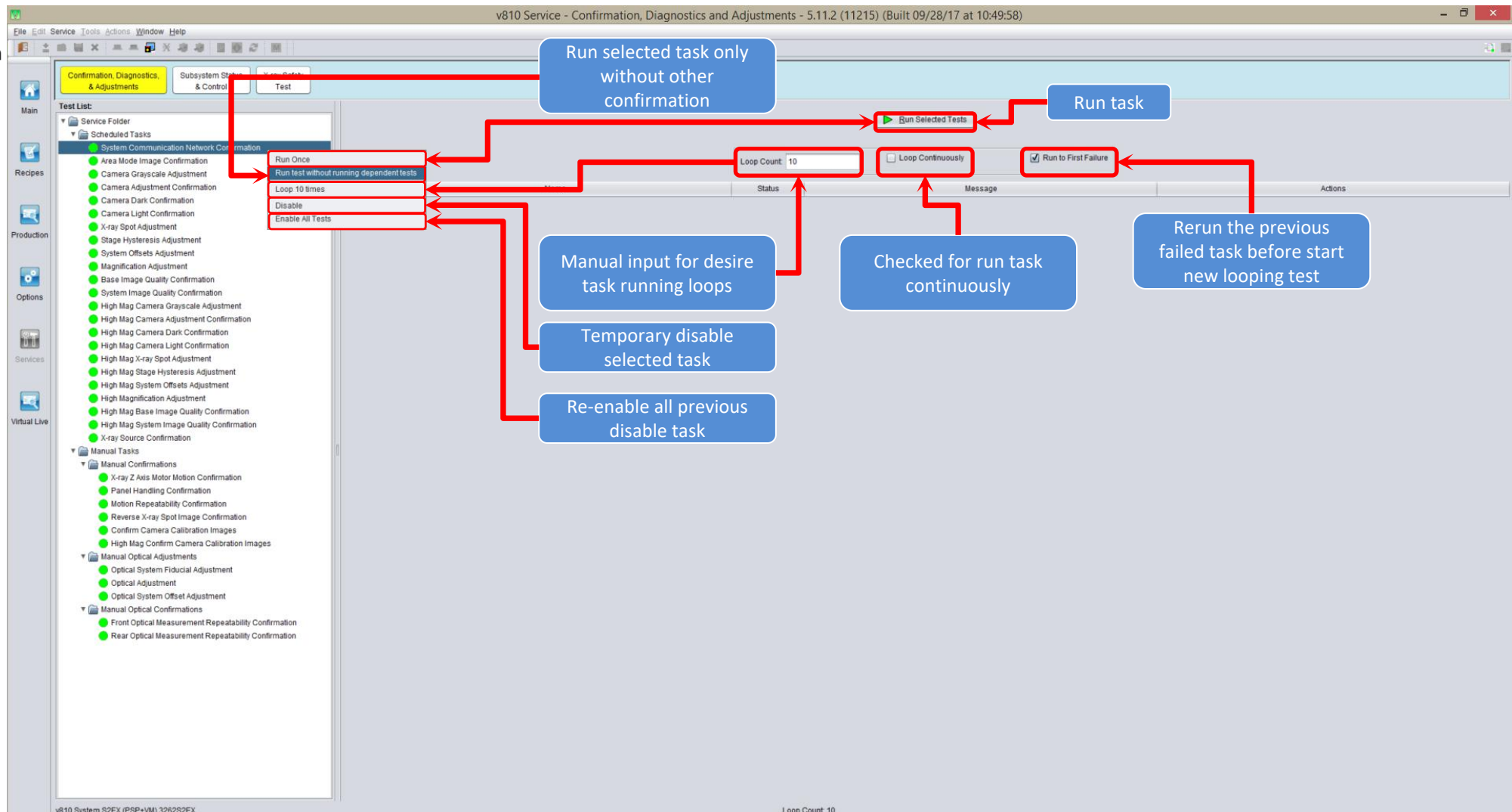
Run Selected Tests

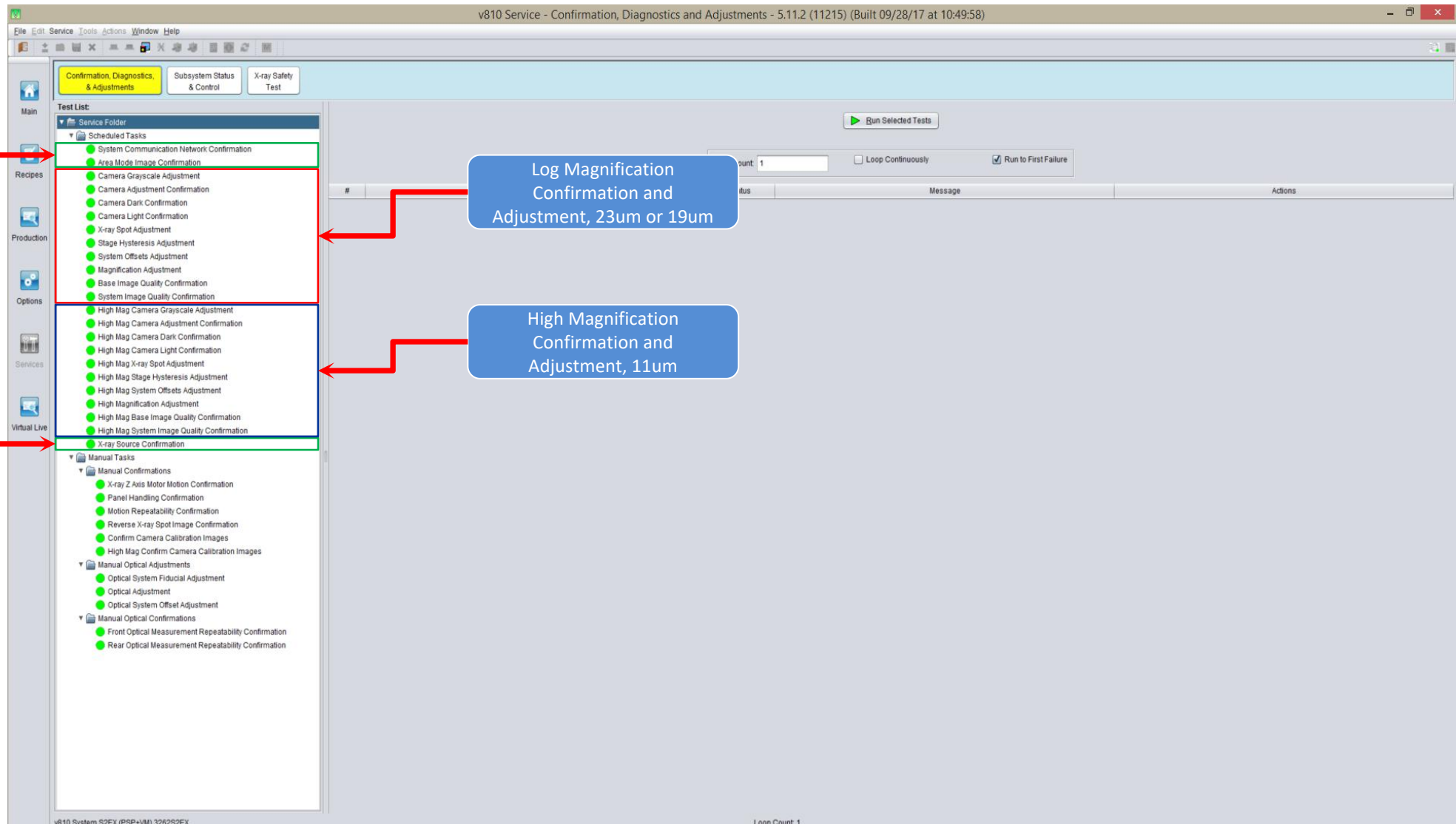
Loop Count: 1 ☐ Loop Continuously ☒ Run to First Failure

#	Name	Status	Message	Actions
---	------	--------	---------	---------

v810 System S2EX (PSP+VM) 3262S2EX

Loop Count: 1







Confirmation, Diagnostics & Adjustments - Low Magnification Define

Automated Board Inspection

Current running Magnification setting

AXI System v810 - Main

System Type: v810 System S2EX (PSP+VM) 3262S2EX
Low Magnification: M19
High Magnification: M11

Software Revision: 5.11.2 (11215) (Built 09/28/17 at 10:49:58)
Defect Package Status: Running

Search Recipe/Variation:

No.	Recipe name	Variation Name	Customer name	Version	Last modified	System Type
1	NEPCON		[not specified]	6.0	Nov 14, 2017 4:04:23 PM	S2EX

Change System Magnification

Warning
In order for the system's new magnification to take effect, the software will need to have a proper shutdown and run a full Confirmation, Diagnostic and Adjustment scheduled tasks after that.

Low Magnification: M19
High Magnification: M11

Restart Now Cancel

Selected Recipe Image

Status: There is 1 recipe available. Auto-pull recipes: OFF

Select desire Low Magnification to use

ABI Important Terms and Files

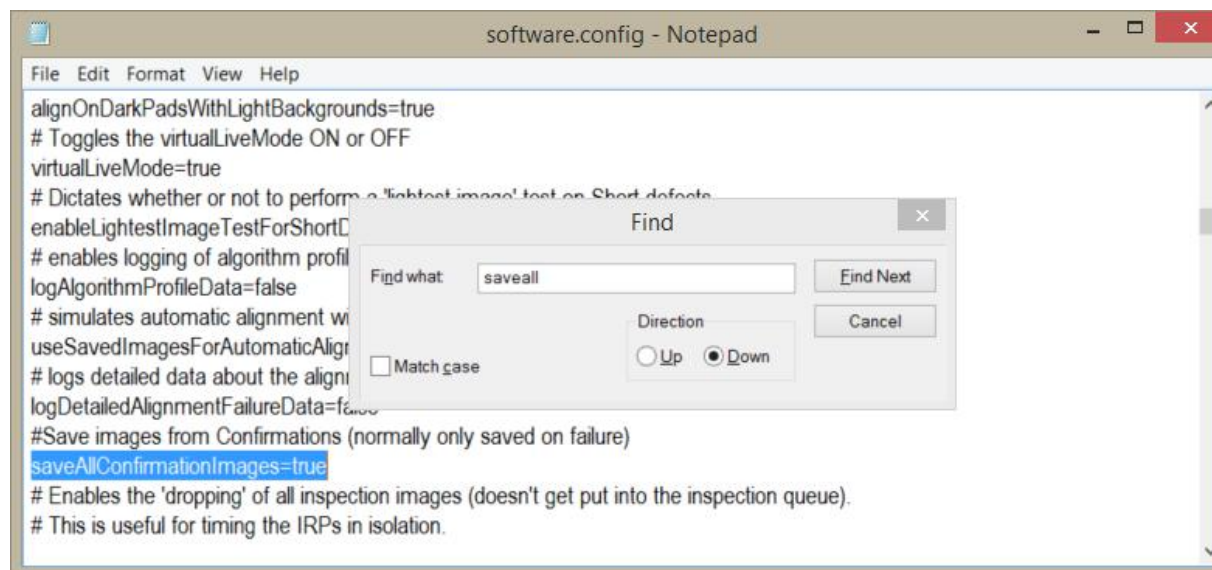
Automated Board Inspection

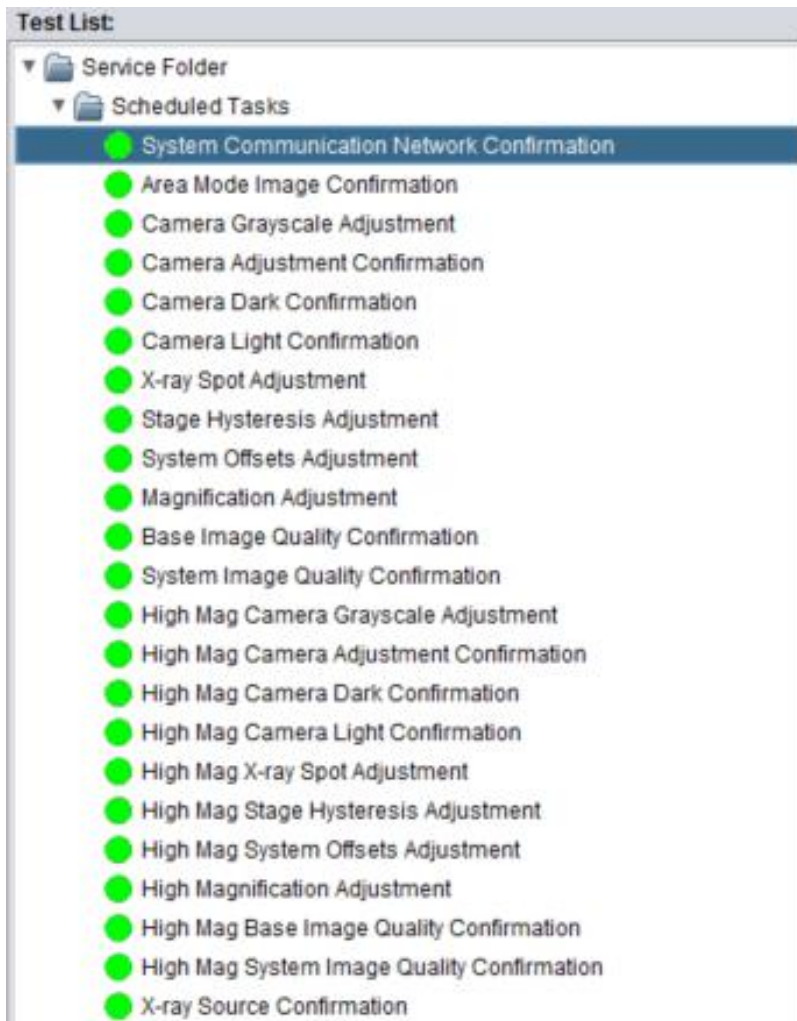
➤ Hardware files

- C:\Program Files\AXI System\V810\<Rev>\calib\hardware.calib
- C:\Program Files\AXI System\V810\<Rev>\config\hardware.config

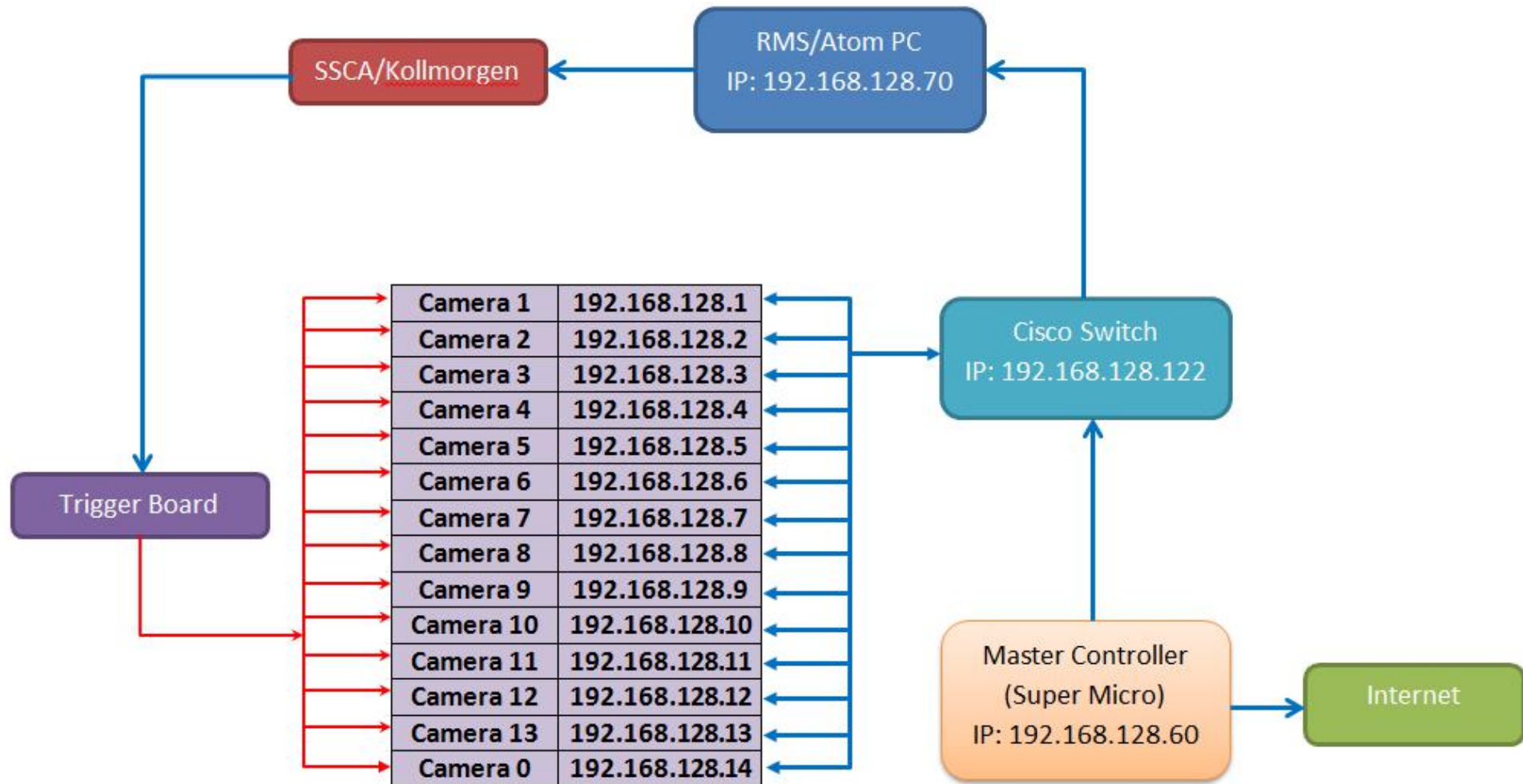
➤ Software files

- C:\Program Files\AXI System\V810\<Rev>\config\software.config
- To log all CD&A tasks related images, user have to enable image logging setting in “Software.config” and relaunch V810i GUI and rerun CD&A tasks to make effect.





- Checks for network communication from the system controller to the IRPs.
- To check for the network connectivity between devices below with the main controller:
 - 4 IRPs,
 - 14 X-Ray cameras
 - Gigabit Switch
 - Remote Motion Server (Atom PC)
- A very simple “ping” test that assures network connectivity exists between all the components in the V810 system. Each specific subsystem also has connection tests that occur when the specific subsystems are initialized (ie. When the first connection to the cameras is initiated).
- Runs at every time machine “Start-up”.
- Log result under path, C:\Program Files\AXI System\v810\<Rev>\log\confirm\telecommunications

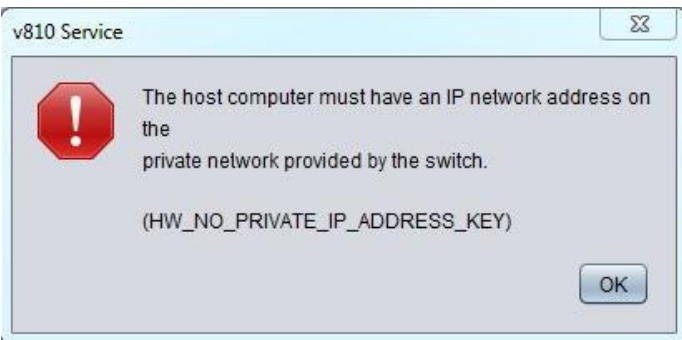




If communication to one or more devices fails:

1. Power checking – Look for lights on failing device.
2. Check cabling.
3. Cycle power and/or reboot the devices.
4. Check the network configuration (IP setting) in the hardware.config file.
5. If hardware failure, replace the failing device.
6. Open “CMD” and ping devices IP, expected feedback from return from ping device.
 - # Remote Motion Server Time-to-Live (TTL) returning should be 128
 - # Each X-Ray cameras Time-to-Live (TTL) returning should be 64
 - 64 is restricted to the same region
 - 128 is restricted to the same continent

Label "Switch"



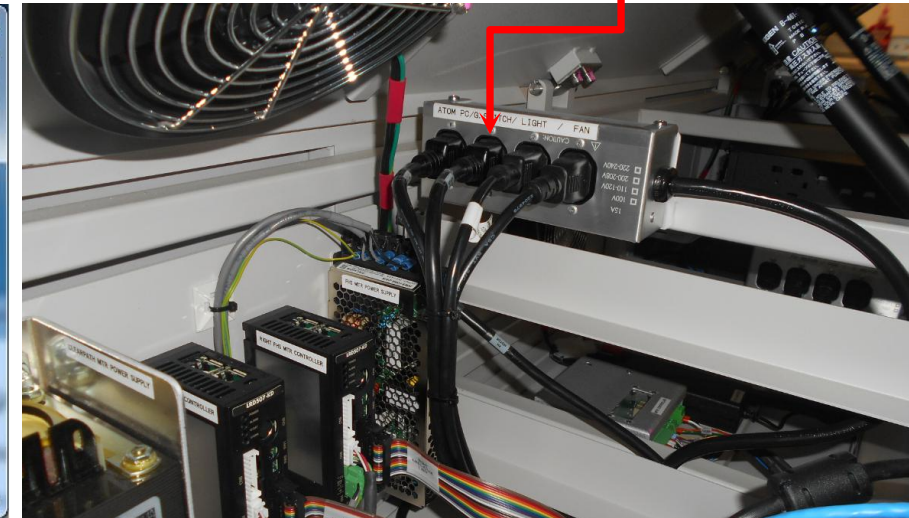
```
Administrator: Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.128.122

Pinging 192.168.128.122 with 32 bytes of data:
Reply from 192.168.128.122: bytes=32 time=1ms TTL=64
Reply from 192.168.128.122: bytes=32 time<1ms TTL=64
Reply from 192.168.128.122: bytes=32 time<1ms TTL=64
Reply from 192.168.128.122: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.128.122:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

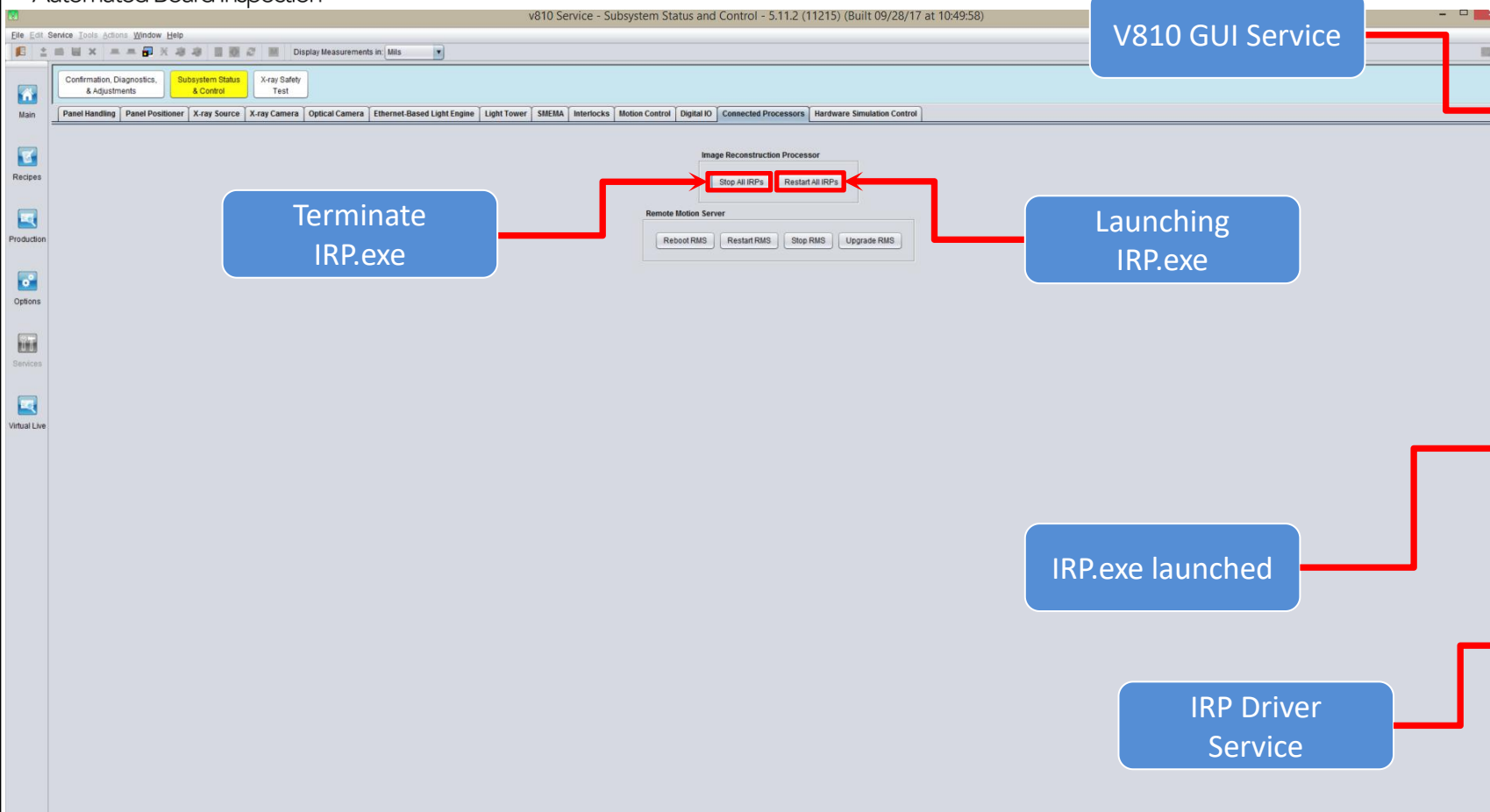
C:\Users\Administrator>
```



- Possible root cause , Gigabit Switch not power up or hardware defective (network card at main controller, LAN cables loosen or broken, or Gigabit switch failed itself)
- Gigabit Switch Time-to-Live (TTL) returning should be 64
- Gigabit Switch itself do not have any hard power cycle button, direct re-seat the power input as power cycle effect

ABI Troubleshooting - Gigabit Switch

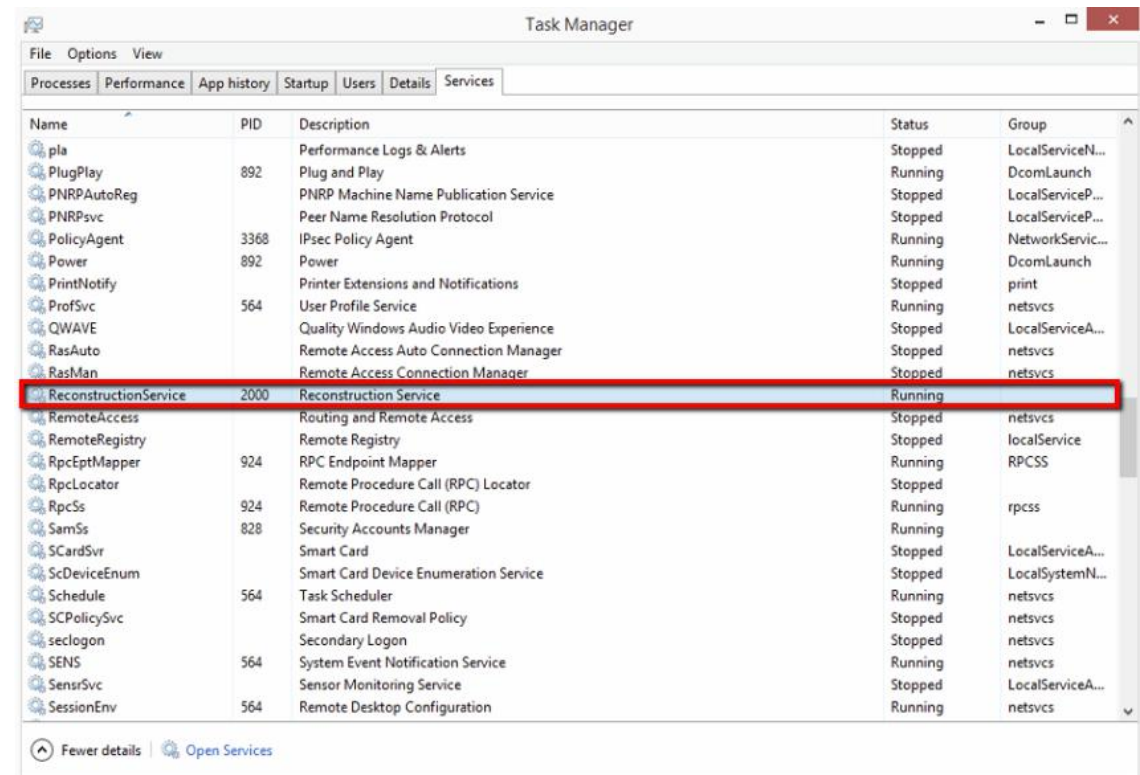
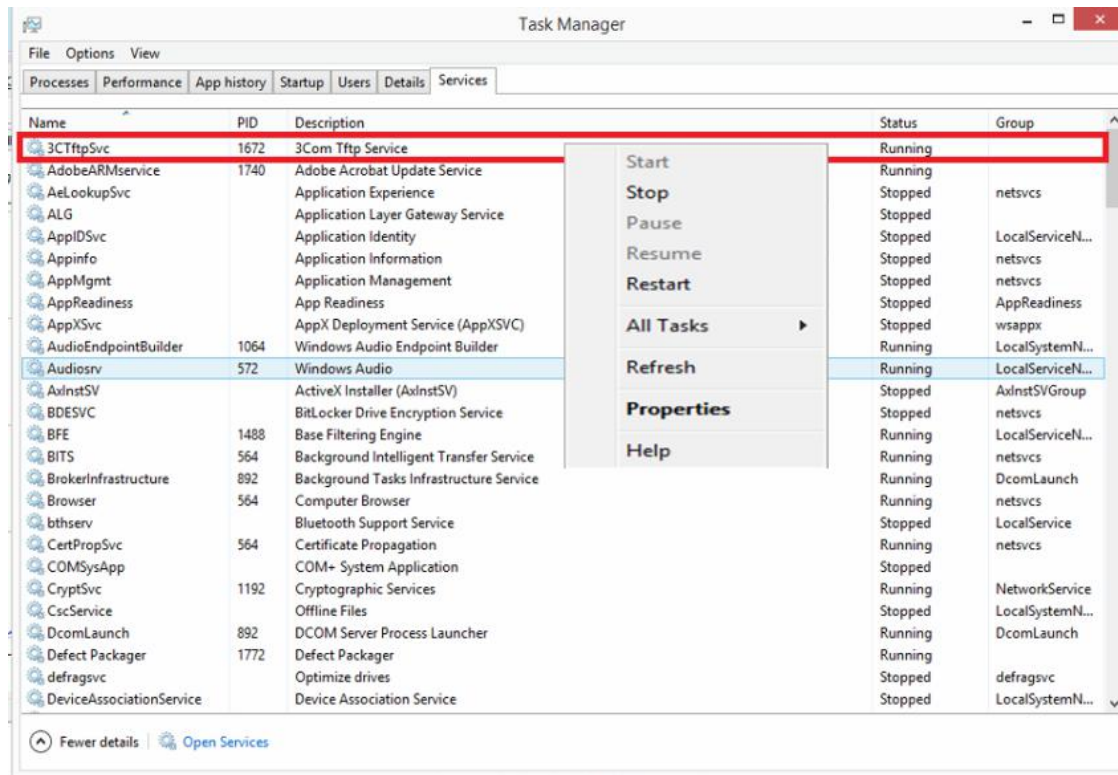
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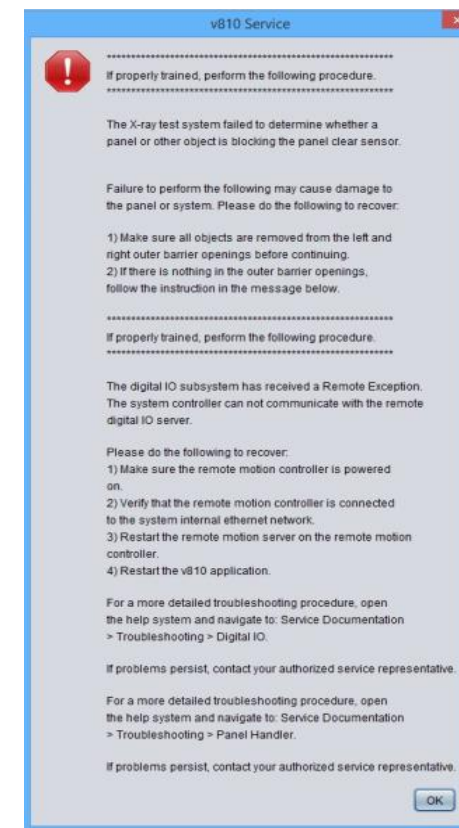
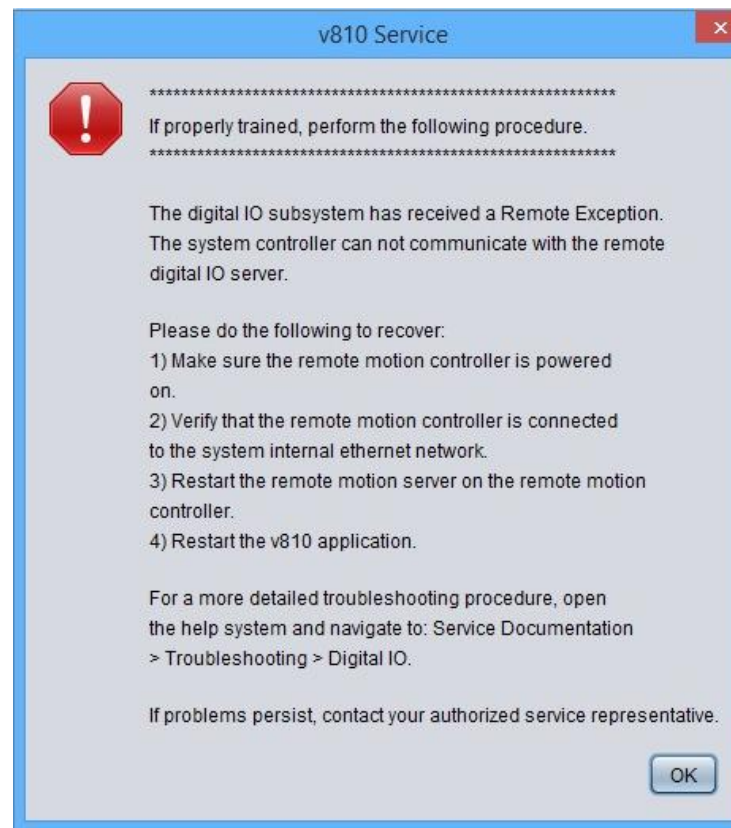
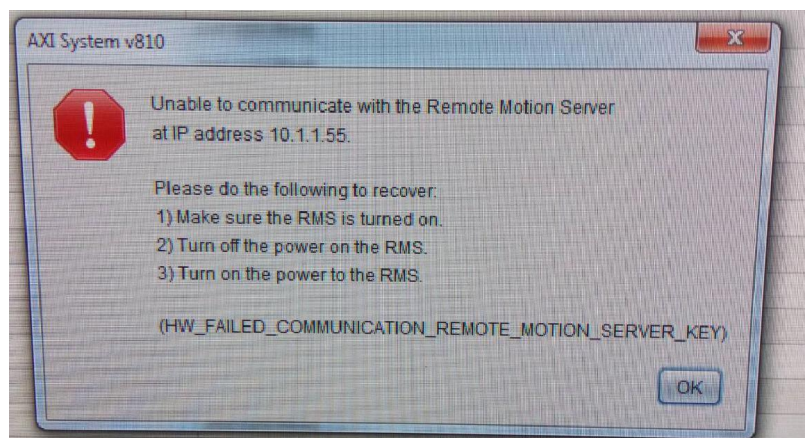
V810i GUI - Connected Processor tab

Name	PID	Status	User name	CPU	Memory (p...	Description
3CTftpSvc.exe	1616	Running	SYSTEM	00	516 K	3CTftpSvc
armsvc.exe	1636	Running	SYSTEM	00	680 K	Adobe Acrobat Update Service
audiodg.exe	20336	Running	LOCAL SE...	00	3,312 K	Windows Audio Device Graph Isolation
cmd.exe	48240	Running	Administr...	00	424 K	Windows Command Processor
conhost.exe	8532	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	740	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	59120	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	59728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	28372	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	50664	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	25744	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	5728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	44272	Running	Administr...	00	532 K	Console Window Host
csrss.exe	576	Running	SYSTEM	00	1,468 K	Client Server Runtime Process
csrss.exe	71412	Running	SYSTEM	00	1,136 K	Client Server Runtime Process
dwm.exe	26620	Running	DWM-1	00	9,352 K	Desktop Window Manager
explorer.exe	29504	Running	Administr...	00	37,328 K	Windows Explorer
IPROSetMonitor.exe	1680	Running	SYSTEM	00	608 K	Intel® PROSet Monitoring Service
IRP.exe	5408	Running	SYSTEM	00	624 K	IRP
IRP.exe	38272	Running	SYSTEM	00	620 K	IRP
IRP.exe	65424	Running	SYSTEM	00	628 K	IRP
IRP.exe	50904	Running	SYSTEM	00	620 K	IRP
IRP.exe	4948	Running	SYSTEM	00	632 K	IRP
IRP.exe	24496	Running	SYSTEM	00	628 K	IRP
IRP.exe	25536	Running	SYSTEM	00	620 K	IRP
IRP.exe	10964	Running	SYSTEM	00	632 K	IRP
javaw.exe	10440	Running	Administr...	00	260,716 K	Java(TM) Platform SE binary
lsass.exe	764	Running	SYSTEM	00	7,084 K	Local Security Authority Process
nvsvs.exe	988	Running	SYSTEM	00	1,716 K	NVIDIA Driver Helper Service, Version 347.25
nvsvs.exe	21568	Running	SYSTEM	00	3,028 K	NVIDIA Driver Helper Service, Version 347.25
nvwm64.exe	148	Running	SYSTEM	00	680 K	NVIDIA WMI Provider
nvwm64.exe	39840	Running	SYSTEM	00	3,204 K	NVIDIA WMI Provider
nvdsync.exe	29620	Running	SYSTEM	00	4,700 K	NVIDIA User Experience Driver Component
perl.exe	4516	Running	Administr...	00	692 K	perl
ReconstructionServi...	1784	Running	SYSTEM	00	3,584 K	ReconstructionService
SearchIndexer.exe	1820	Running	SYSTEM	00	30,804 K	Microsoft Windows Search Indexer
services.exe	752	Running	SYSTEM	00	4,304 K	Services and Controller app
smss.exe	436	Running	SYSTEM	00	216 K	Windows Session Manager
spoolsv.exe	1468	Running	SYSTEM	00	2,240 K	Spooler SubSystem App
svchost.exe	832	Running	SYSTEM	00	6,184 K	Host Process for Windows Services
svchost.exe	880	Running	NETWORK...	00	6,964 K	Host Process for Windows Services
svchost.exe	468	Running	LOCAL SE...	00	13,084 K	Host Process for Windows Services
svchost.exe	804	Running	SYSTEM	00	26,924 K	Host Process for Windows Services

Task Manager



- Image Reconstruct Processor driver installed in main controller.
- System Communication Network Confirmation is checking and ensure the IRP.exe and services in activated.
- IRP driver services tasks can manual activate as below



- Possible root cause, window or RMS driver corrupted, hardware failure (Atom PC or connectivity error (LAN with Gigabit Switch or Gigabit Switch LAN port))

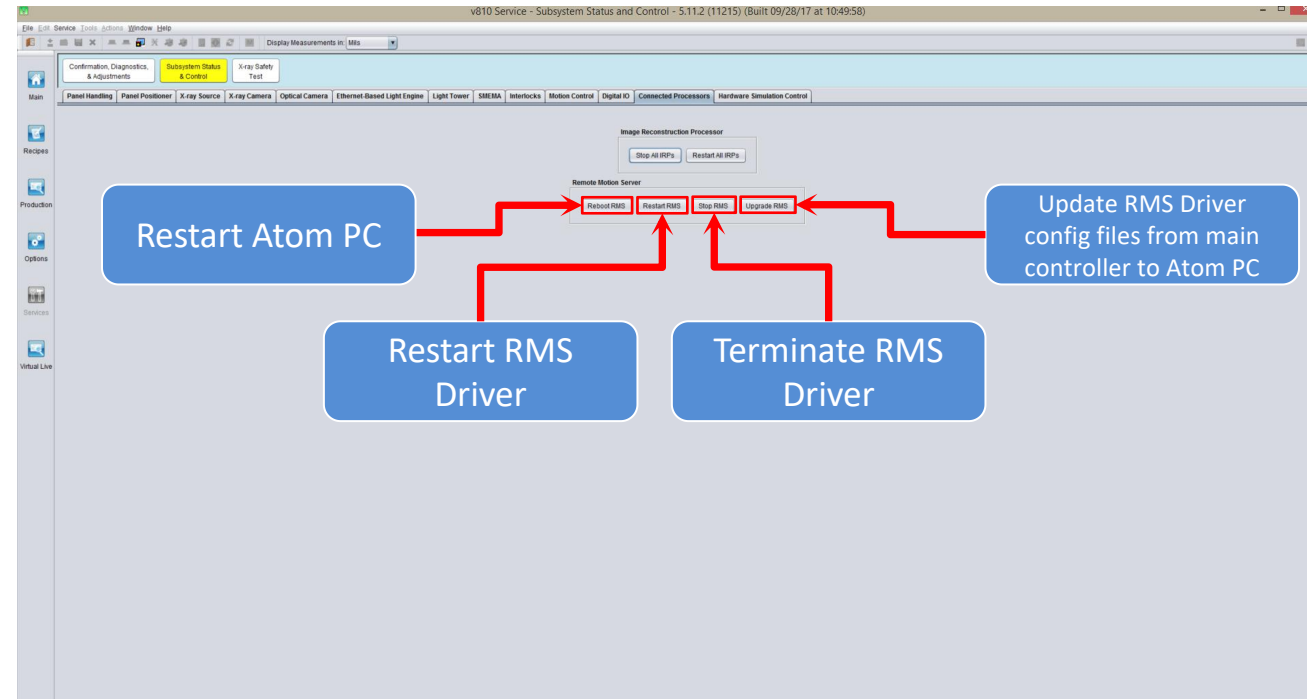
```
Administrator: Command Prompt
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.128.70

Pinging 192.168.128.70 with 32 bytes of data:
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.128.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

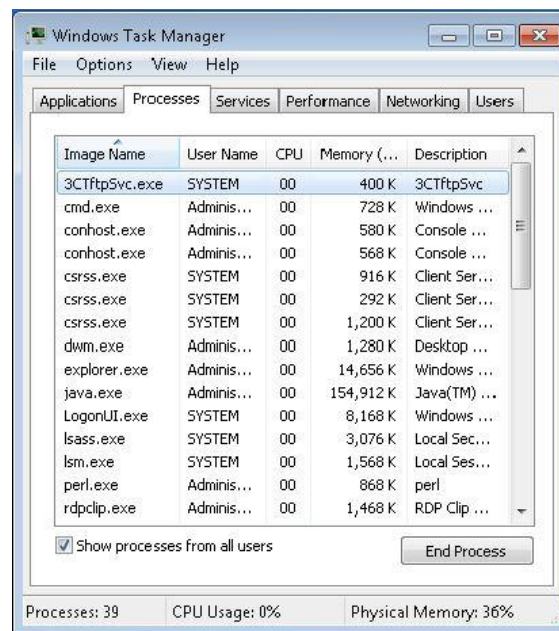
C:\Users\Administrator>
```



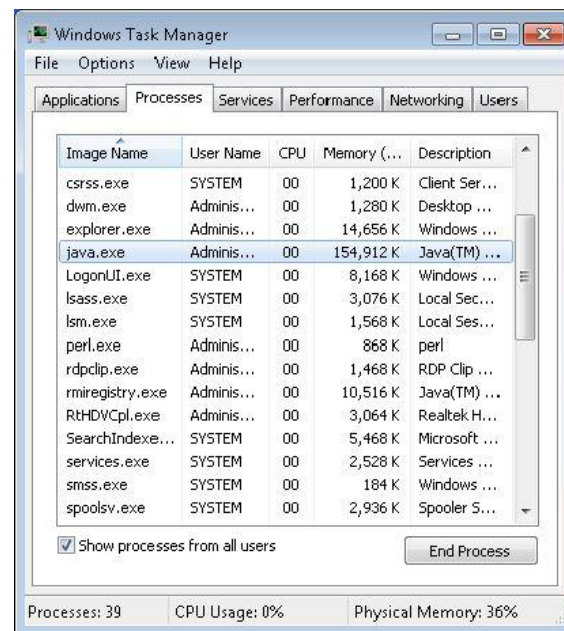
- RMS Driver, a driver to allow V810i GUI communicate with SSCA via Atom PC.
- Remote Motion Server Time-to-Live (TTL) returning should be 128



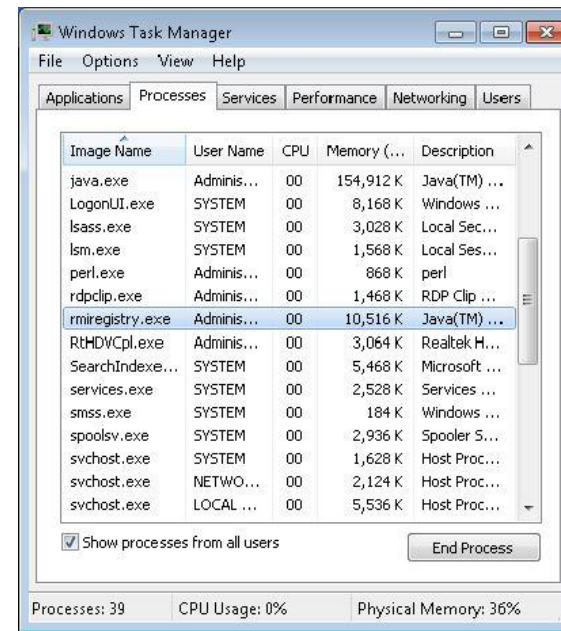
Remote Desktop Access



3CTftpSvc.exe

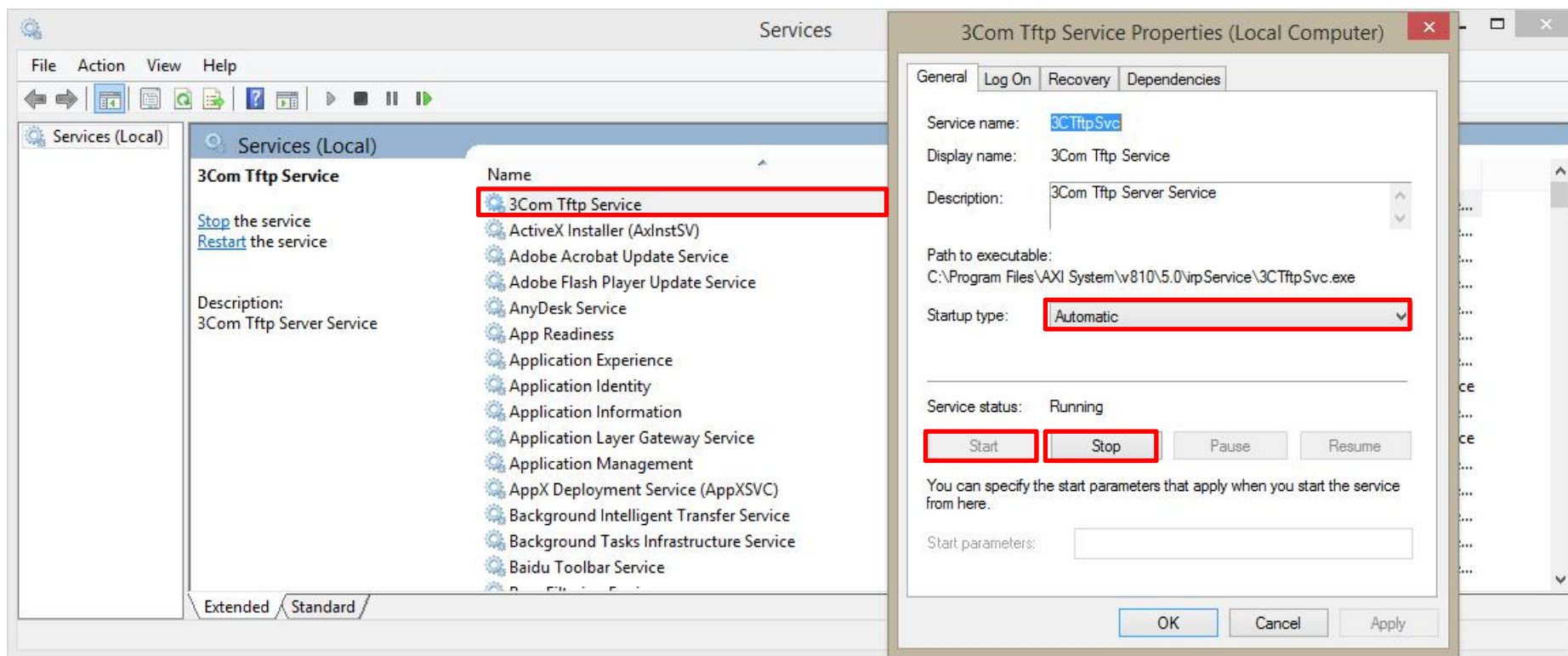


java.exe

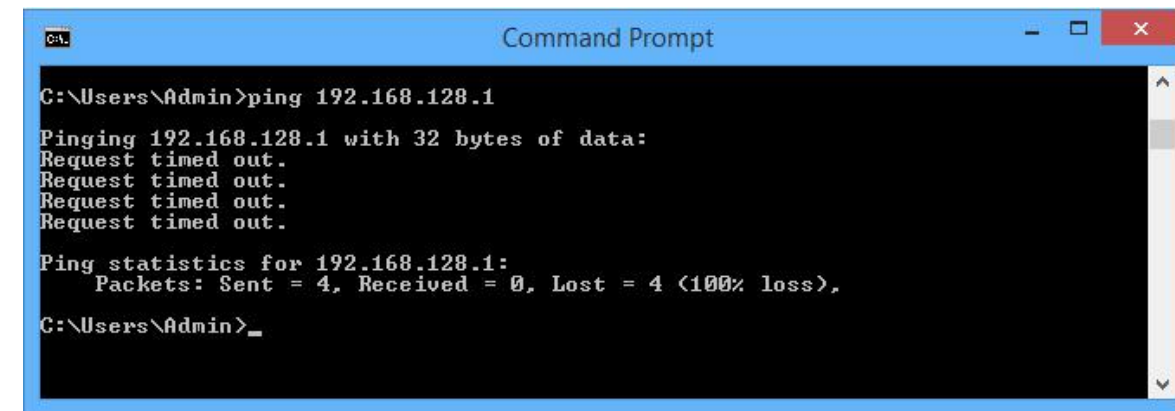
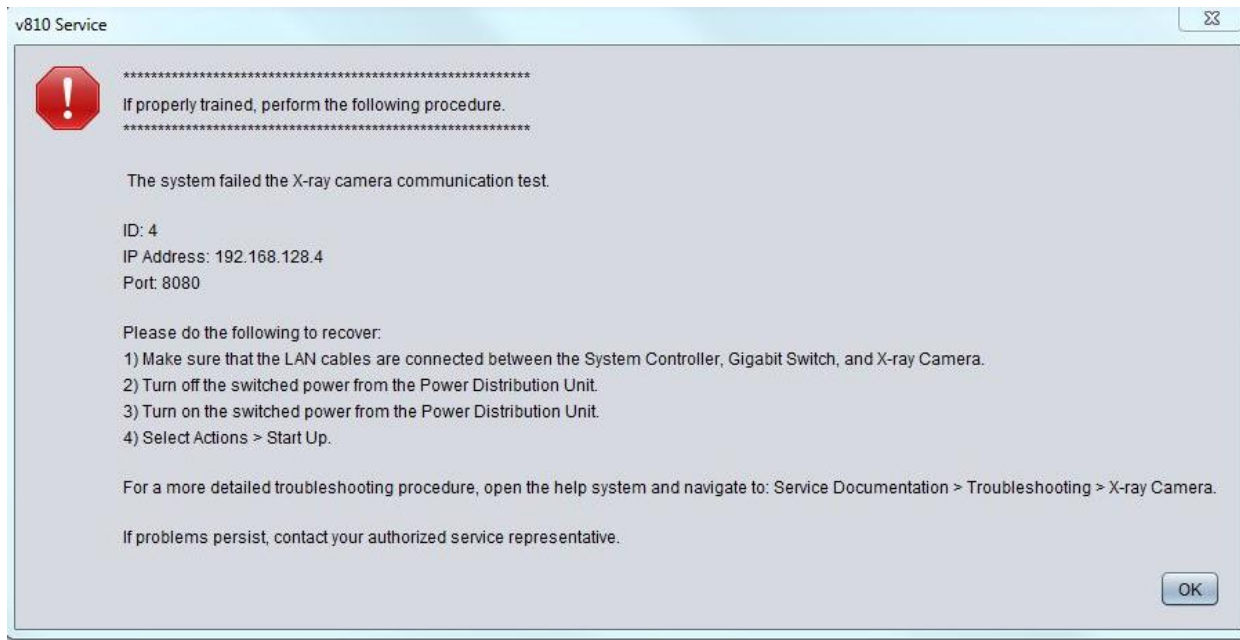


rmiregistry.exe

- Once launching RMS driver at V810i GUI, 3 exe should also launching at Atom PC
- Java.exe will only visible once RMS drive launch without error



- If **3CTftpSvc.exe** and **rmiregistry.exe** not able to initialize, go to services and refresh (Stop and Start) the both **3CTftpSvc** and **rmiregistry** service and ensure the Start-up type is **Automatic**.

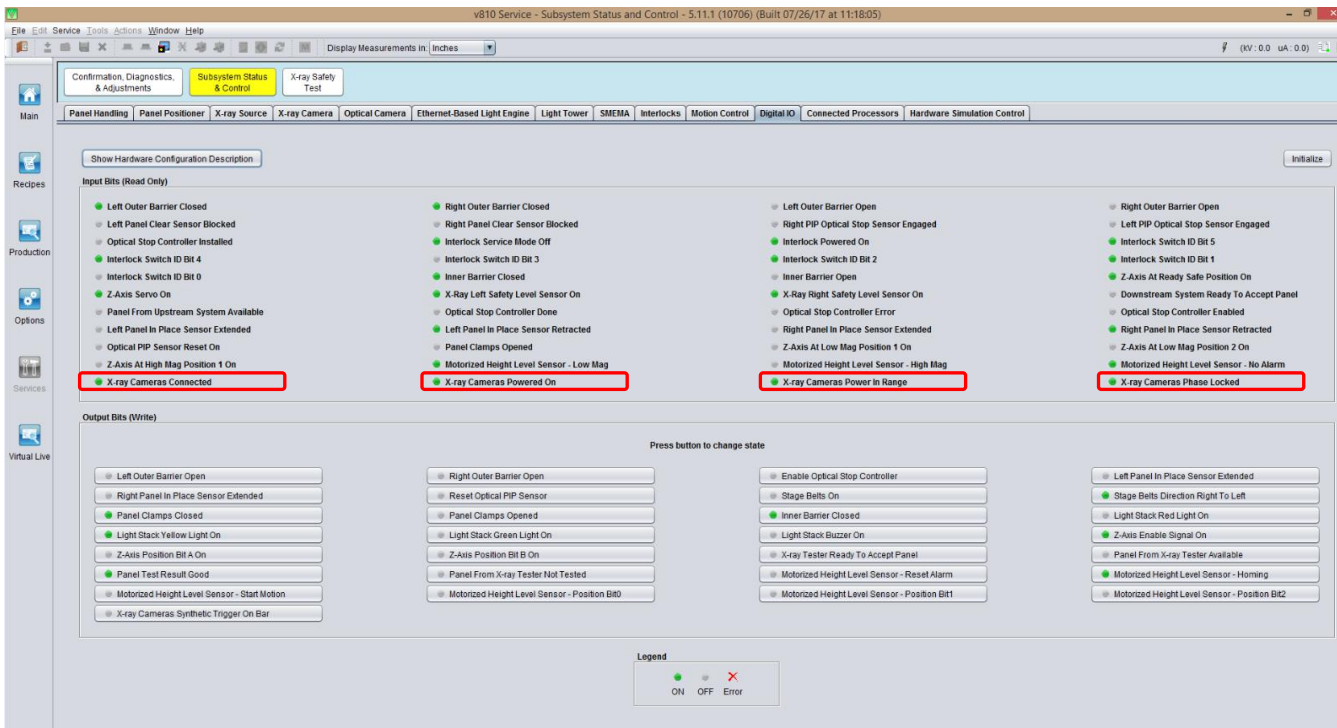


- Possible root cause, connectivity error, dust blocking IR sensor, hardware failure, window firewall activated, trigger board not power on.
- If system configured with fix jumper camera IP, do not have dusty problem
- X-Ray camera Time-to-Live (TTL) returning should be 64



Troubleshooting - X-Ray Camera Communication Error

Automated Board Inspection



Power cycle
trigger board



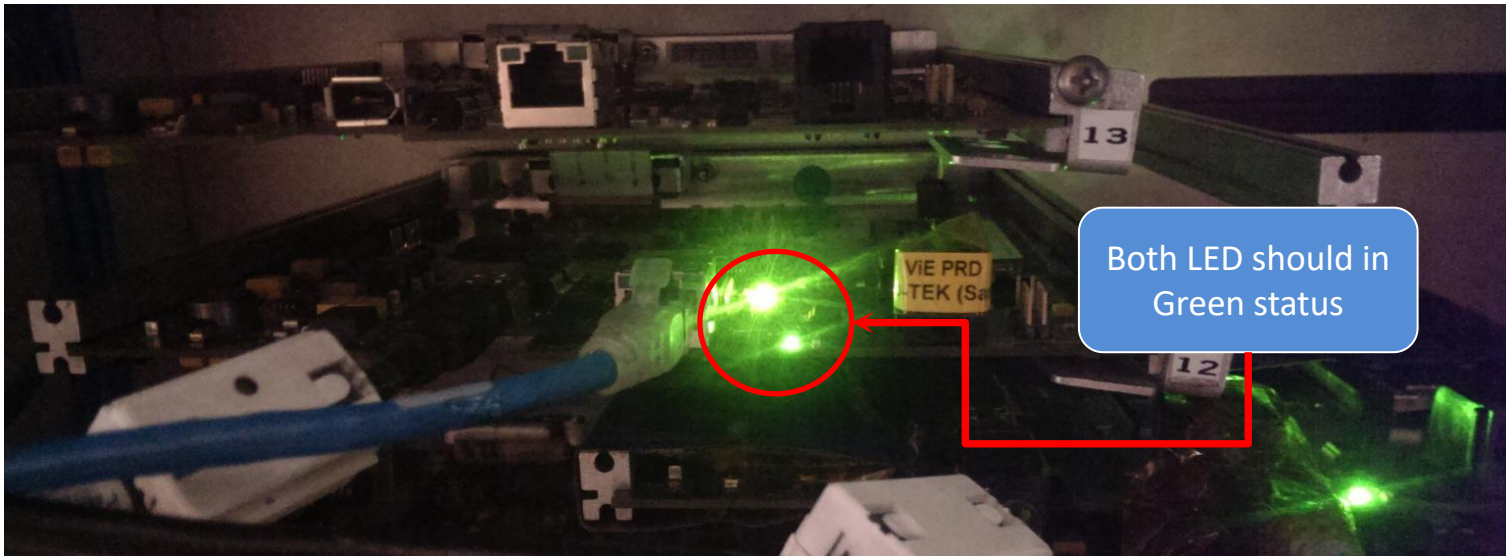
➤ Power cycle Trigger board assembly and checked at DIO tab, once trigger board powered, the DIO should display as shown



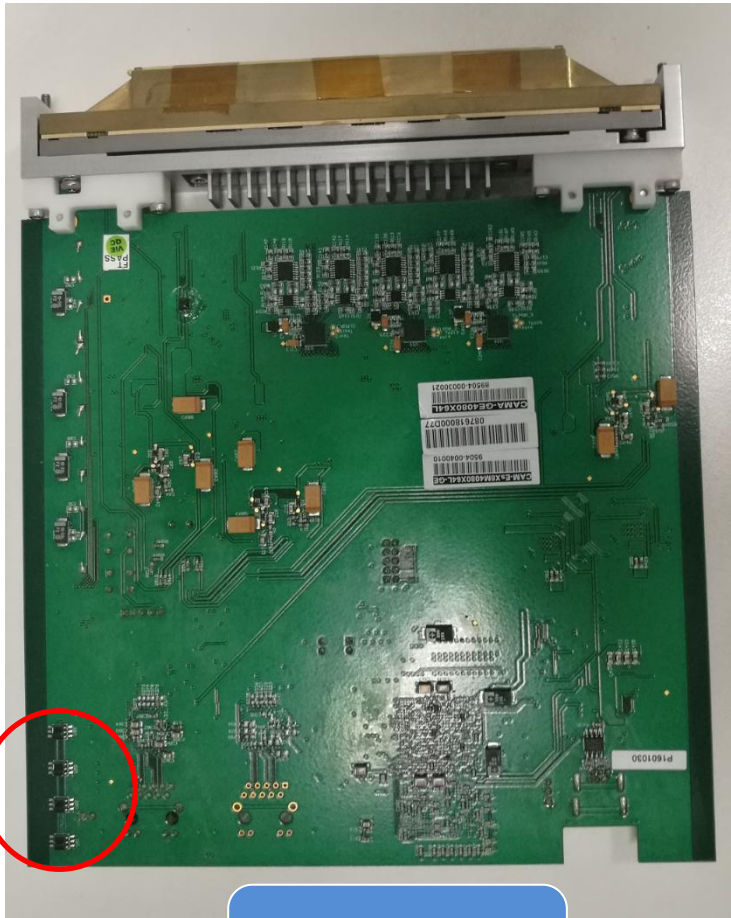
Troubleshooting - X-Ray Camera Communication Error

Automated Board Inspection

4mm distance
between Flag and
IR sensor

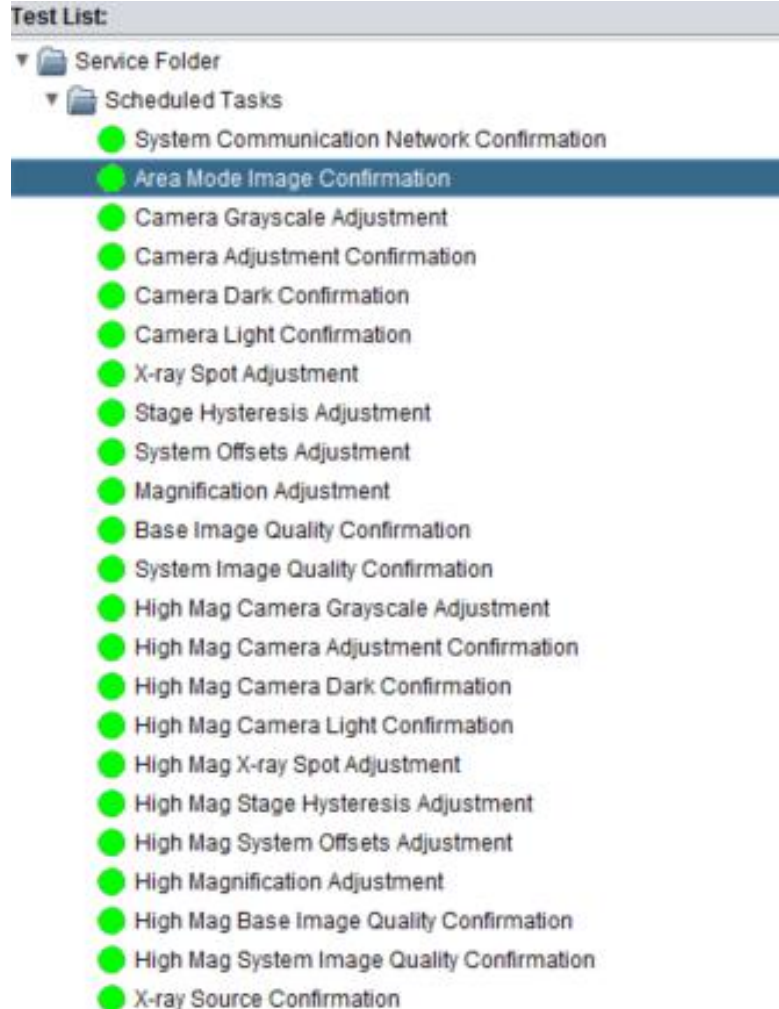


Both LED should in
Green status



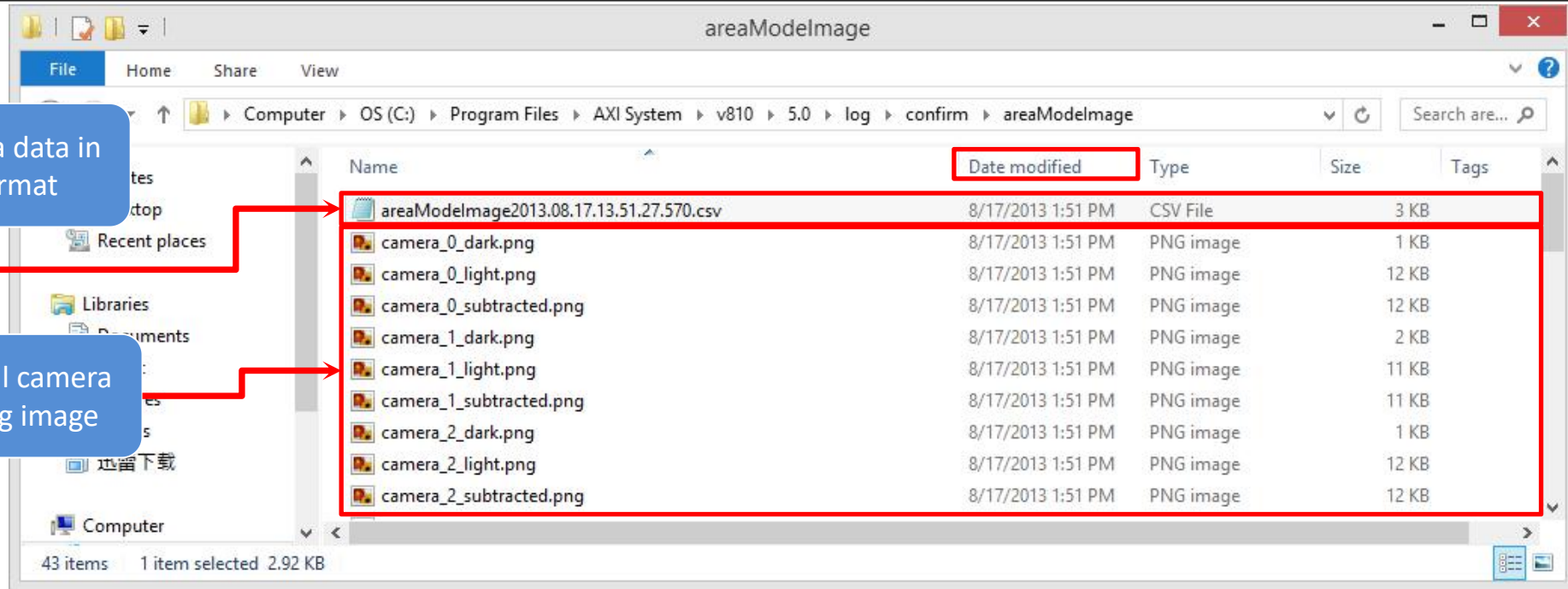
If dusty, clear with
IPA if needed

Automated Board Inspection

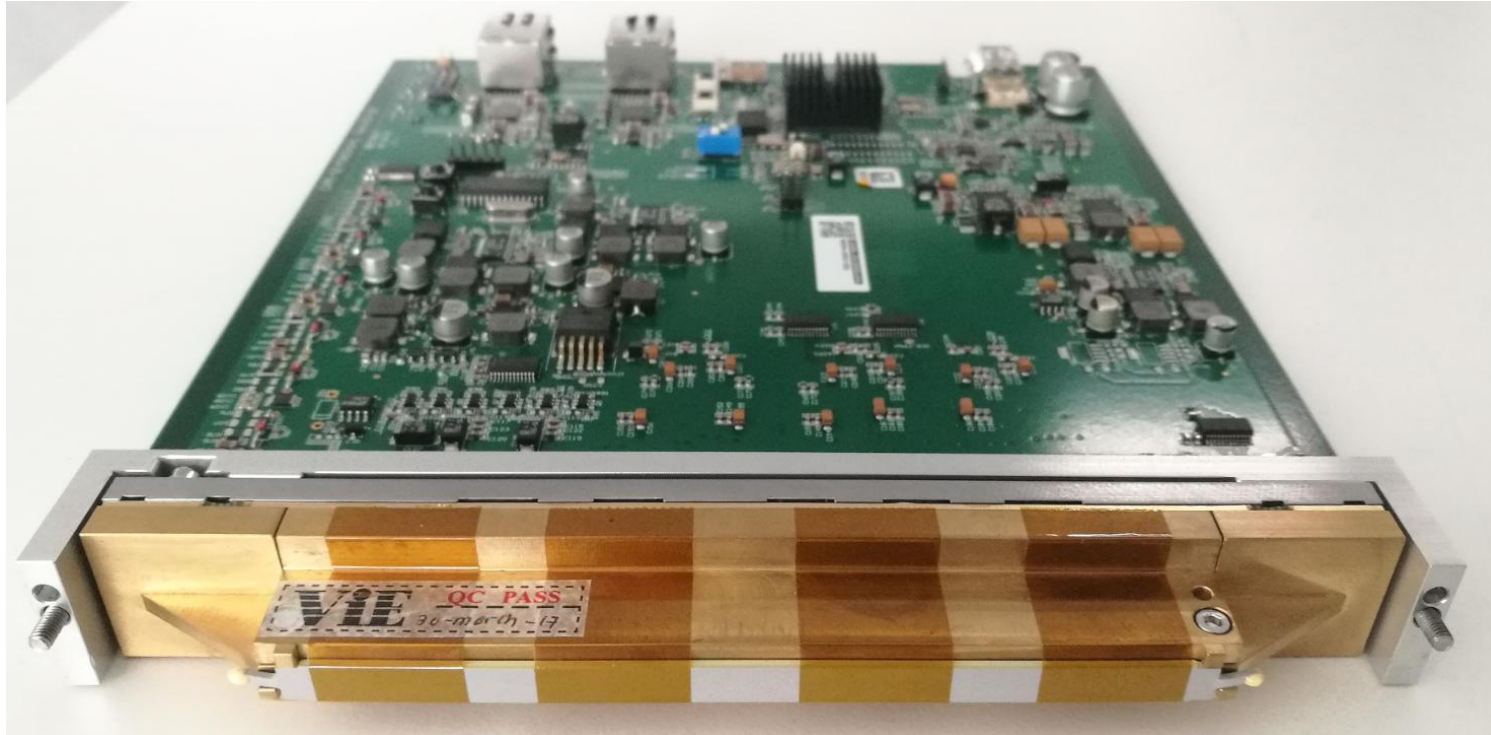


- Checks for a reasonable area mode image on each camera.
- Main purpose is to verify
 - Make sure that there is no debris/component/tools drop on top of the cameras
 - Make sure no bad segment on camera's scintillator.
- During this task, Inner barrier shutter plate must open a
- Loaded PCB panel need to be unload before confirmation start.

Automated Board Inspection

All camera data in
CSV formatIndividual camera
snapping image

- Logs result to directory "C:\Program Files\AXI System\v810\<REV>\log\confirm\areamodelImage "
- Validate result log file with latest date modified.



camera_XX_dark

camera_XX_subtracted

camera_XX_light



Sample 1: Bad segment and camera degrading



Sample 2: Full segment bad



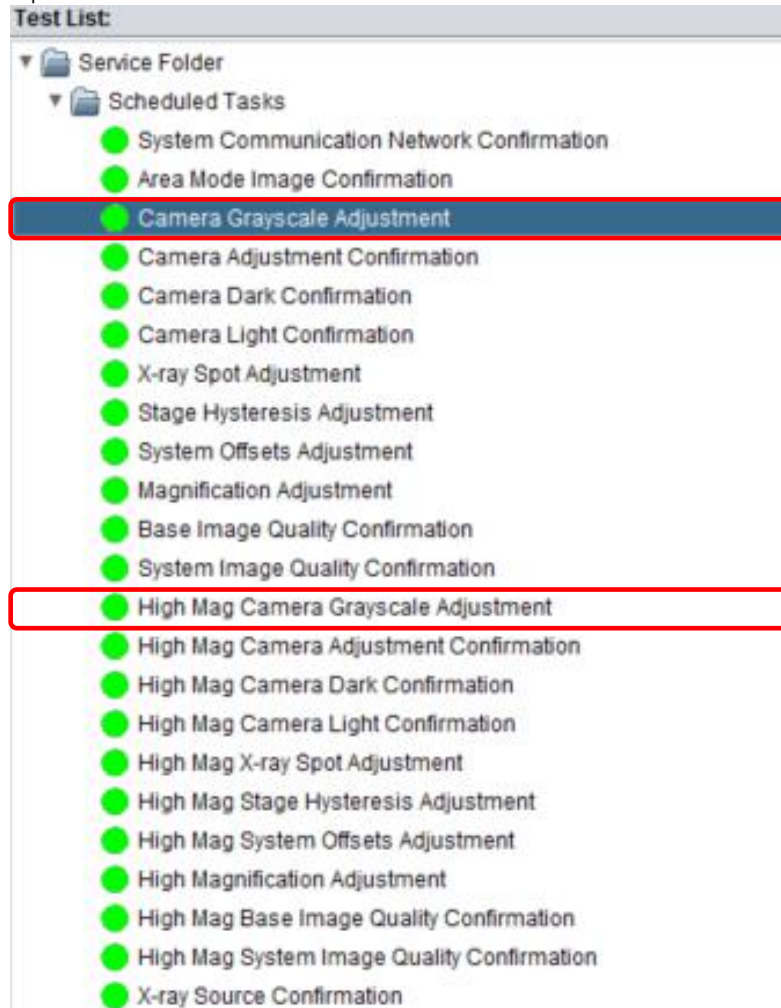
Sample 3: Debris on top of Camera Field Stop Assembly

Troubleshooting:

1. Check for any debris on any foreign material on the top of Field Stop Assembly
2. Ensure inner barrier in "Open" position
3. Swap the error camera with other camera on machine to identical if the camera defective

ABI Schedule Tasks - Camera Grayscale Adjustment

Automated Board Inspection



- Runs at start-up and panel load & unload if it's been more than 10 minutes.
- Cannot have a panel loaded in stage, will impact final result.
- Uses synthetic triggers and area mode images.
- It's a self calibration on segment and pixel gain and offset for each camera.



Troubleshooting - Camera Grayscale Adjustment

Automated Board Inspection

Confirmation, Diagnostics, & Adjustments

Subsystem Status & Control

X-ray Safety Test

Test List:

- Service Folder
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 - System Communication Network Confir
 - Area Mode Image Confirmation
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 - Camera Dark Confirmation
 - Camera Light Confirmation
 - X-ray Spot Adjustment
 - Stage Hysteresis Adjustment
 - System Offsets Adjustment
 - Magnification Adjustment
 - Base Image Quality Confirmation
 - System Image Quality Confirmation
 - X-ray Source Confirmation

Run Selected Tests

Loop Count: 1

Loop Continuously

Run to First Failure

#	Name	Status	Message	Actions
1	Connection to IRPs Confirmation	Passed		
2	Connection to RMS Confirmation	Passed		
3	Area Mode Image Confirmation	Passed		
4	Camera Grayscale Adjustment	FAILED	Camera(s) 7 FAILED*****If prope... Check for debris on camera array.	

Sample error message

AXI System v810

!

If properly trained, perform the following procedure.

The Camera Grayscale Adjustment FAILED

Camera(s) 13,12,11,10,9,8,7,6,5,4,3,2,1,0 FAILED

If properly trained, perform the following procedure.

The X-ray camera detected an invalid segment gain result during grayscale adjustment.

ID: 13
IP Address: 192.168.128.13
Port: 8090
Segment number: 0
Result: 1000.0
Low limit: 1.0
High limit: 5.0

Please do the following to recover:
1) Check for debris on the camera.

For a more detailed troubleshooting procedure, open the help system and navigate to: Service Documentation > Troubleshooting > X-ray Camera.

If problems persist, contact your authorized service representative.

Check for debris on camera array.

If problems persist, contact your authorized service representative.

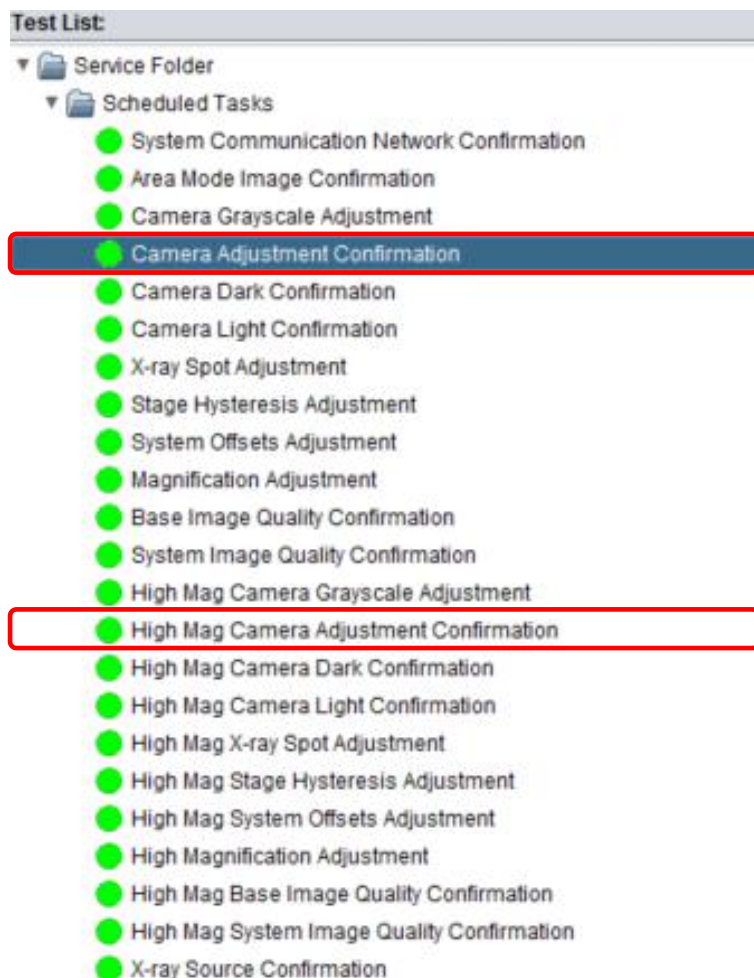
OK

Sample error message

If one or two camera fail:

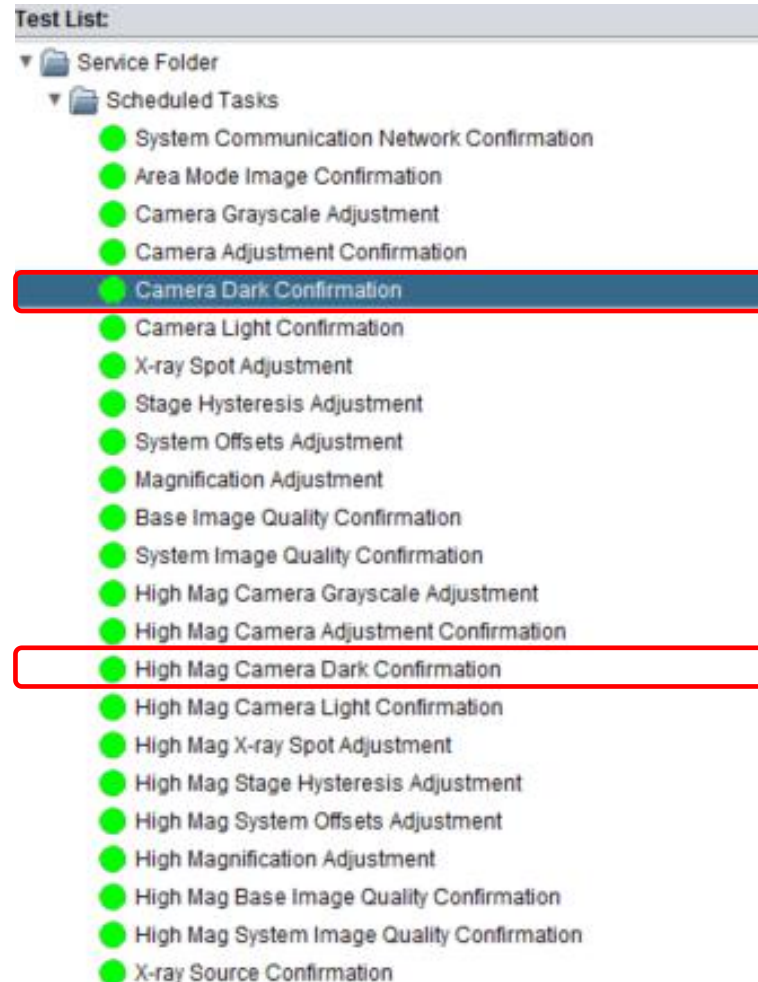
- Check failing devices Area Mode Image
- Replace the failing device

Automated Board Inspection



➤ To obtain the camera segment and pixel gain and offset values so that error could be detected by software earlier before hardware error occur.

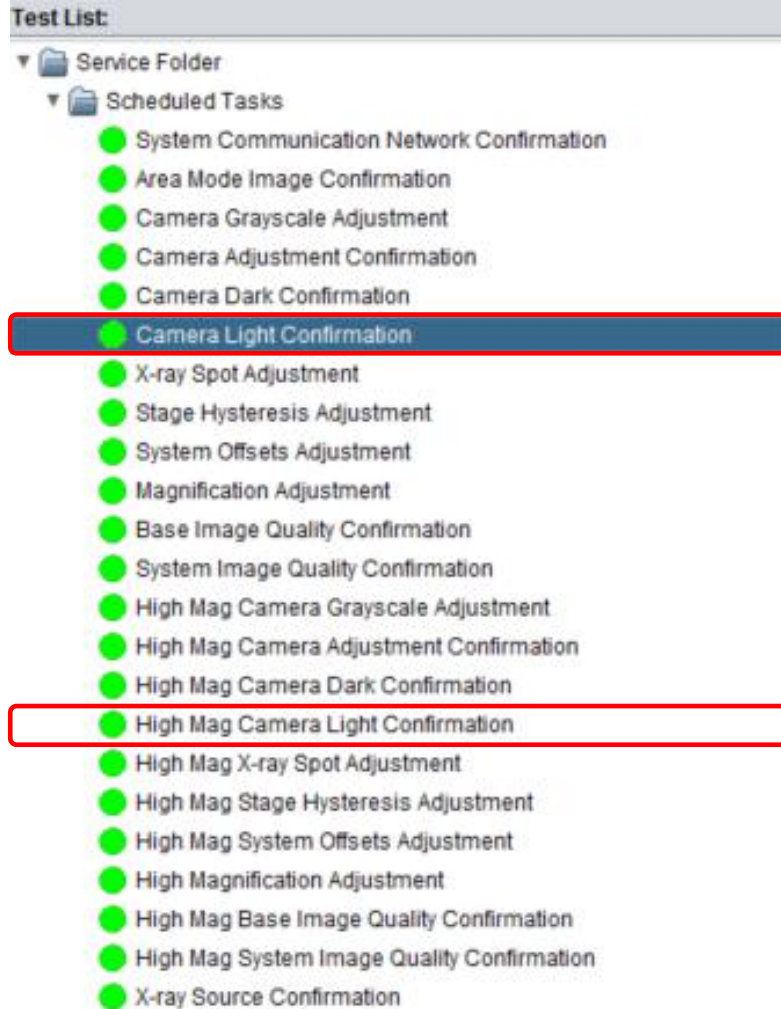
Automated Board Inspection



- To confirm that the camera greyscale, dark value, works and the images are smooth in both X and Y.
- Helps detect noise in the system, X-Ray spot oscillation or failed greyscale calibrations.
- Runs at Start-up and Panel Load & Unload if more than 15 minutes have passed.

ABI Schedule Tasks - Camera Light Confirmation

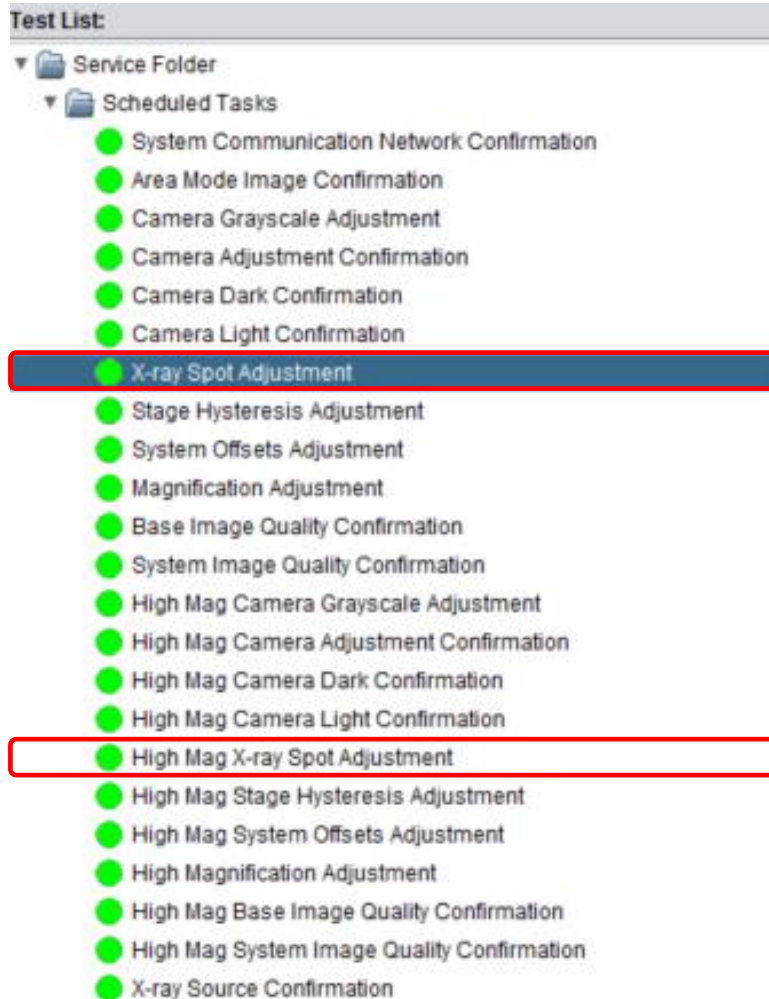
Automated Board Inspection



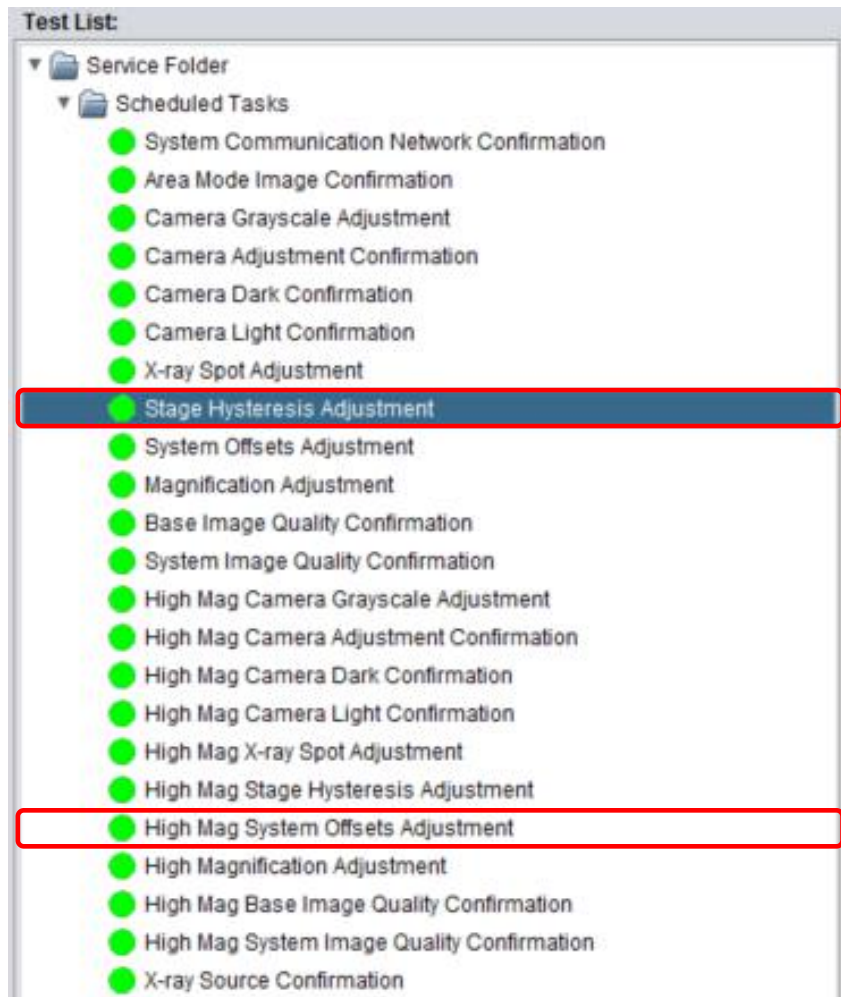
- To confirm that the camera greyscale, light value, works and the images are smooth in both X and Y. This helps detect noise in the system, X-Ray spot oscillation or failed greyscale calibrations.
- Gathers projection images using synthetic triggers with x-ray current on.
- Runs at Start-up and Panel Load if more than 30 minutes have passed.

ABI Schedule Tasks - X-Ray Spot Adjustment

Automated Board Inspection

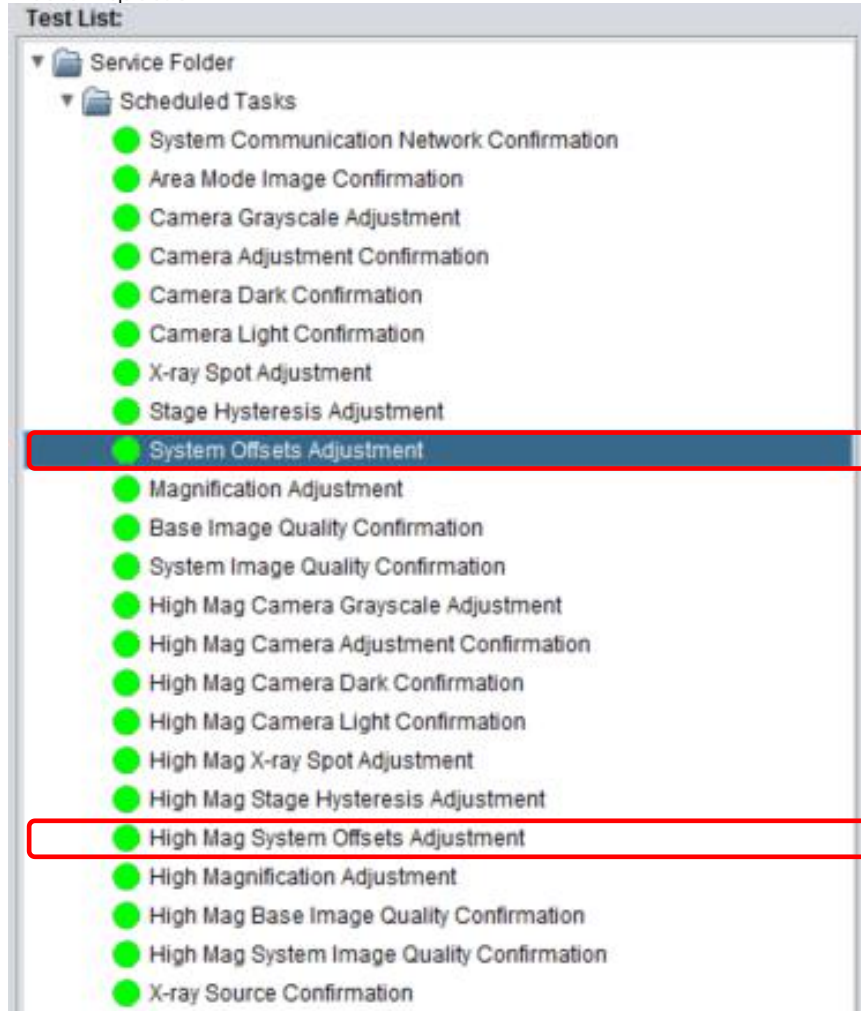


- Calculates the X/Y stage position in relationship to the X-Ray spot via the coupon over the **Camera 0** and determines the required X/Y stage offset to ensure proper stage positioning.
- Calculates the XY position of the stage with the **Fiducial Coupon** centered over the camera array.
- Runs at Start-up, panel load, and panel unload if it's been more than 24 hours.
- Uses motion triggers and searches the image for the system fiducial.



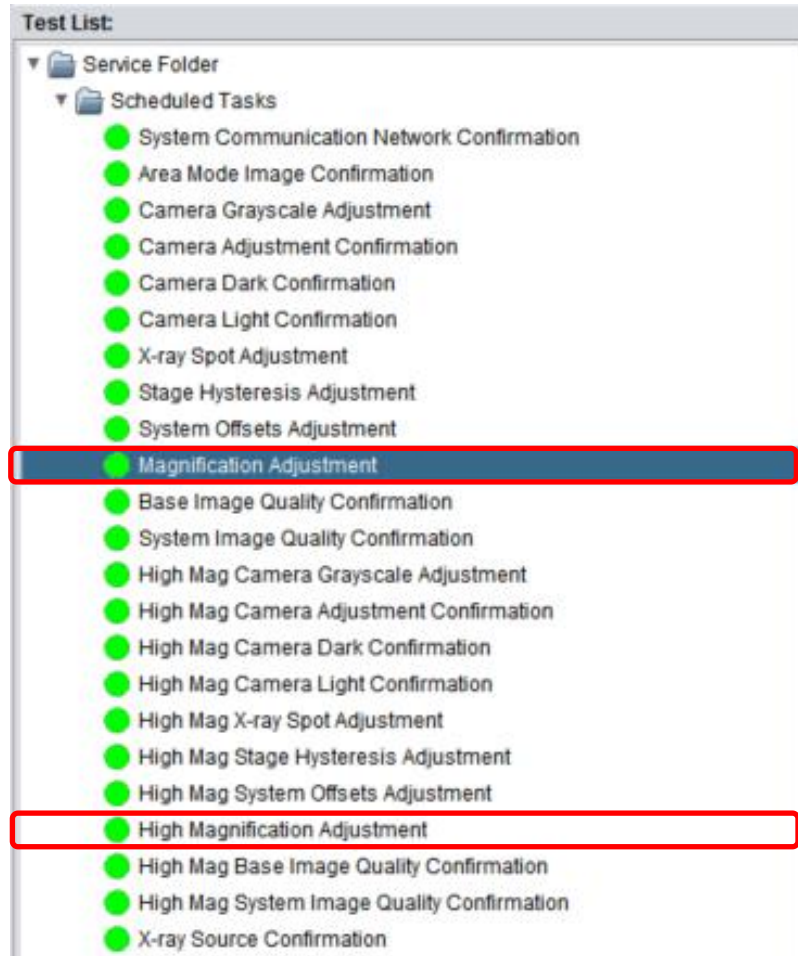
- Adjusts for differences in Y-Axis motion between forward and reverse for image reconstruction. Minimize the offset gap for Y axis movement front and reverse action.
- Runs at Start-up, Panel Load, and Panel Unload if it's been more than 24 hours.

Automated Board Inspection

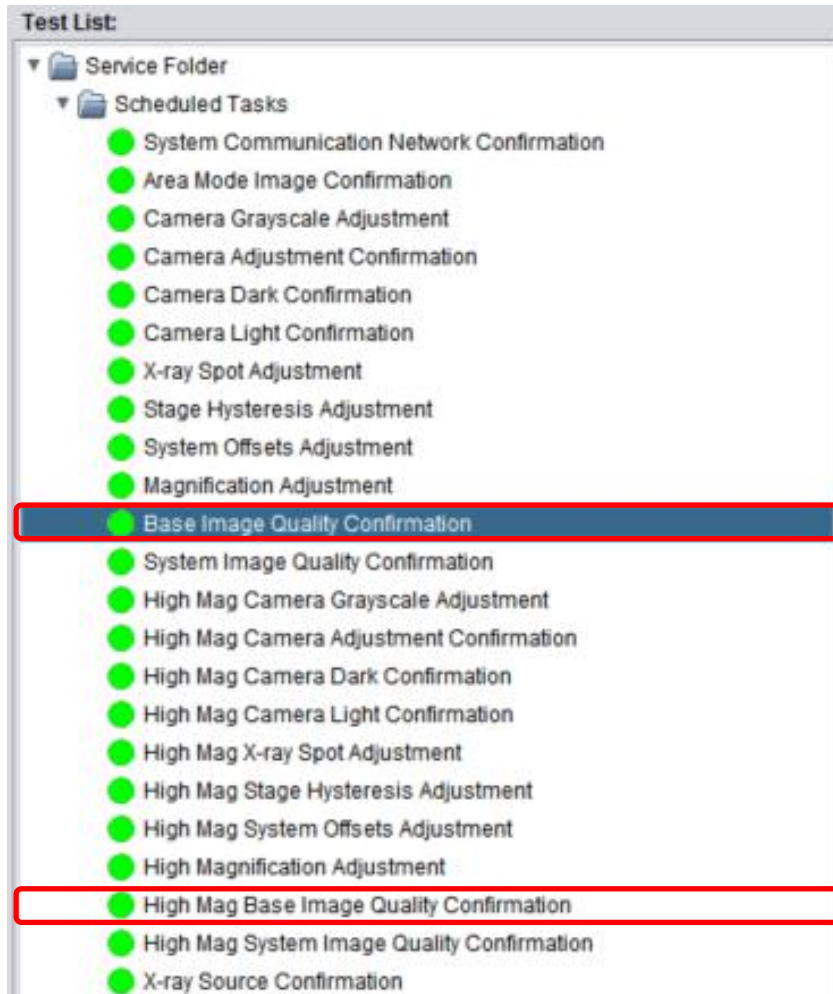


- Determines the stage X/Y position from the center camera to each of the other cameras for image reconstruction. The value is only used during reconstruction. It is passed to the IRP's and is used to do a final image alignment during the reconstruction process.
- Runs at Start-up, Panel Load, and Panel Unload if it's been more than 24 hours.

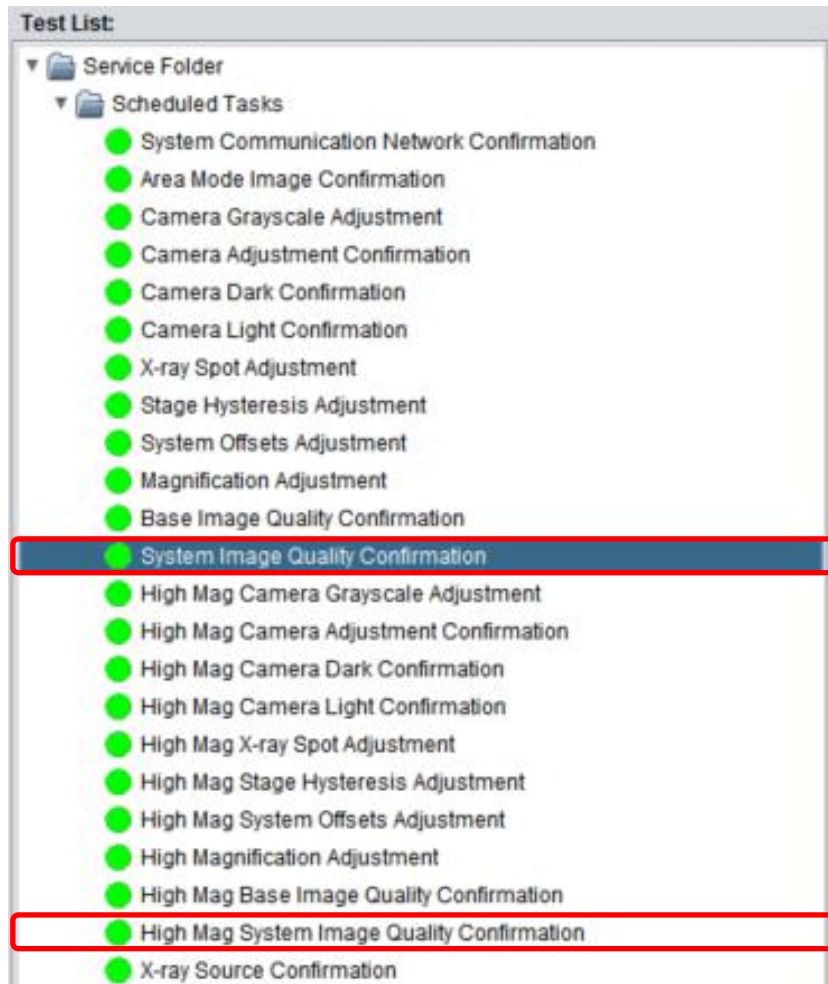
Automated Board Inspection



- Determines the Reference Plane magnification. By design the reference plane
- Runs at Start-up, Panel Load, and Panel Unload if it's been more than 24 hours.
- Generates reconstructed images of the vertical edge on the Fixed Rail Right Fiducial triangle using different values for Magnification and determines which value generates sharpest images.



- Determines Image Quality by generating projection images of the quadrilateral on the **Fiducial coupon**.
- Runs at Start-up if it has been more than 24 hours since the last time.
- Averages 16 images over the center camera.
- If this confirmation fails, it logs the last failing image.



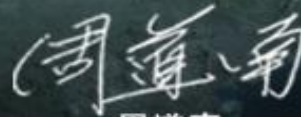
- Determines system level Image Quality.
- Most comprehensive Image Quality check. Involves all subsystems.
- Runs at Start-up, Panel Load, and Panel Unload if it's been more than 30 minutes.
- Generates reconstructed images of the Quadrilateral on the **Fiducial Coupon**. This coupon is at a different Z height than the Fixed Rail Right Fiducial Coupon.
- Trigger **14 cameras** to snap image, reconstruct and diagnostic.



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